

2110576 Agentic AI for Computer Engineering

2110576 · Presentation-Based Practice · 3 Credits · Offered both semesters

Department of Computer Engineering · Chulalongkorn University

Course Information

Field	Detail
Course Code	2110576
Course Title	Agentic AI for Computer Engineering
Credits	3 Credits (3-0-9) — one 3-hour session per week, 9 hours self-study
Level	Graduate
Format	Presentation-based practice course — no lectures examined; students learn topics at home and present results in class. No midterm or final examination.
Prerequisite	Basic programming (Python); no prior AI background assumed
Duration	Sat 8 Aug – 28 Nov: 1 orientation + 12 topic weeks (Phase 1) + a 4-session final project (Phase 2), ending with the final pitch on 28 Nov
Semester	Offered both Semester 1 and Semester 2 (Saturday section)
Target Students	Computer Science and Computer Engineering students (CM and CS) - Saturday 13:00 - 16:00 for part-time (weekend) program - Wednesday 13:00 - 16:00 for full-time (weekday) program*

***Note:** The full-time (weekday) section is conducted at a more intensive pace alongside the CP program. Batches 1 and 2 cover different sets of topics, while Batch 3 focuses on agent harness implementation rather than additional advanced topics. The final project emphasizes agent system design, implementation, and performance evaluation, rather than merely demonstrating a use case.

Course Description

Agentic AI for Computer Engineering is a hands-on, presentation-driven course on building and working with agentic AI systems using Claude — from prompting and the API, through tool use and basic agents, to advanced architectures, the Model Context Protocol (MCP), multi-agent orchestration, the Claude Agent SDK, evaluation, safety, and deployment. The course consolidates the former *Claude Basic* and *Claude Advance* special-topics courses into a single progressive track.

Rather than sitting through lectures and written exams, students **learn by doing**. The semester has two phases. In **Phase 1**, working in fixed groups of three that share one Claude account, students pick a topic from the batch released for the current round (Basic → Intermediate → Advanced), practise it at home, and present their results in a booked week; groups not presenting write homework on the same assigned topics, and every session ends with structured note-taking and discussion. Across Phase 1 the class works through **36 topics** spanning the basic-to-advanced spectrum. In **Phase 2**, each group runs a **final project** — defining a use case in a workshop, giving a progress update, and pitching a finished solution to invited guests.

Learning Outcomes

Upon successful completion, students will be able to:

- Explain how large language models and Claude work, and choose appropriately among Claude's agentic products (API/SDK, the no-code Claude Cowork desktop agent, and the developer-focused Claude Code).

- Write structured prompts, call Claude through the Messages API and SDK, and produce validated structured output.
- Define tools, implement the tool-use loop, and build agents with memory and retrieval.
- Apply a disciplined Claude Code and Claude Cowork workflow — Plan Mode, the Explore→Plan→Code→Test→Commit loop, a lean CLAUDE.md, context budgeting, hooks, and skills.
- Specify an agent's identity and behaviour (SOUL.md, CLAUDE.md) and design advanced architectures using planning, reflection, and the ReAct pattern.
- Use and build MCP servers, build multi-agent systems, and build production agents with the Claude Agent SDK.
- Design and run agent evaluations, and apply safety, guardrail, cost, and deployment practices.
- Present technical work clearly, critically evaluate and challenge AI output, and contribute substantively to peer discussion.

Class Organization & Accounts

The class is organized into **12 fixed groups of 3 students** (about **36 students**). All practice — presentations, homework, and discussion — is done and graded on a group basis and stays with the same group for the whole semester.

The course provides **12 shared Claude accounts — one per group**. Each group manages and coordinates its own account (taking turns, tracking message/usage limits, avoiding conflicts) so the allocation lasts the full semester.

How the Course Works

Two phases: topic rounds, then a final project

The semester has two phases. **Phase 1 (Rounds 1–3)** is topic-based: three rounds of four weeks, **3 groups presenting per week**, each round releasing a **batch of 12 topics** at a higher level of difficulty. Every group presents **once per round — 3 topic presentations** in all. **Phase 2 (Round 4)** is a **four-session final project**: a use-case workshop, two progress-update weeks, and a final pitch with invited guests.

Round	Weeks	Format	Level / focus
Orientation	1	Grouping, rules, accounts, first topic brief	—
Round 1	2–5	3 topic presentations/week	Batch 1 — Basic : LLMs, prompting, API, tool use, basic agents
Round 2	6–9	3 topic presentations/week	Batch 2 — Intermediate : Claude Code/Cowork workflow, context engineering, skills, MCP intro
Round 3	10–13	3 topic presentations/week	Batch 3 — Advanced : architectures, MCP servers, multi-agent, Agent SDK, evaluation
Round 4	Project (4 sessions)	Workshop → progress x2 → final pitch	Final project — a real use case built and pitched

Each Phase-1 batch contains **12 distinct topics** — one per group. Every presenting group in a week receives a **different topic**, and across a round all 12 groups cover different topics. Round 4 is described under **The Final Project** below.

The weekly rhythm — Phase 1 (3-hour session)

Every Phase-1 topic week (Weeks 2–13) follows the same structure:

Time	Segment	What happens
First 30 min	Class management	Attendance check; homework check (homework was submitted before class); announcements
Next 2 hours	Presentations	The 3 booked groups present their practice results, one after another (~40 min each including Q&A). All groups take notes and join the discussion.
Last 30 min	Topic assignment & brief	The 3 groups booked for next week receive their individual topics (each different) and a topic brief. This also tells every other group what to write homework on.

*(Round 4 sessions use their own formats — see **The Final Project**.)*

Who does what each Phase-1 week

- **The 3 presenting groups** demonstrate the results they practised at home on their assigned topics.
- **The other 9 groups** each submit **homework** on that week's three presented topics — prepared during the previous week and **submitted before class**.
- **Every student individually** fills a **paper discussion form for each of the three presentations** (one per topic), handed back **at the end of class**. This is **individual work**; students in the presenting groups fill them too. Forms are written in English, kept short, and must be legible — they are **scanned and evaluated with Claude Cowork**, then **double-checked by the lecturer**.

Because a group presents 3 times across the 12 Phase-1 weeks, it writes homework on the other **9** weeks — and every student produces a discussion note in all **12** Phase-1 weeks.

Attendance is required to earn credit. A student who is absent receives **zero for that session's presentation and discussion note** — including a member of a presenting group who is absent on presentation day.

Deliverable deadlines (Phase 1)

Deliverable	Who	Deadline
Presentation	The 3 booked groups	In class, in the booked week
Homework	The 9 non-presenting groups	Before class begins
Discussion forms (3 on paper, one per topic)	Every student individually (presenters included)	Handed in at the end of class (attendance required)

Weekly Schedule

Wk	Date	Phase / Round	Activity	Presenting
1	Sat, 8 Aug	Orientation	Grouping (3/group), rules, account setup; brief Round 1 topics	—
2	Sat, 15 Aug	R1 · Basic	3 presentations + discussion; brief next groups	Groups 1–3

Wk	Date	Phase / Round	Activity	Presenting
3	Sat, 22 Aug	R1 · Basic	3 presentations + discussion; brief next groups	Groups 4–6
4	Sat, 29 Aug	R1 · Basic	3 presentations + discussion; brief next groups	Groups 7–9
5	Sat, 5 Sep	R1 · Basic	3 presentations + discussion; brief Round 2	Groups 10–12
6	Sat, 12 Sep	R2 · Intermediate	3 presentations + discussion; brief next groups	Groups 1–3
7	Sat, 19 Sep	R2 · Intermediate	3 presentations + discussion; brief next groups	Groups 4–6
8	Sat, 26 Sep	R2 · Intermediate	3 presentations + discussion; brief next groups	Groups 7–9
9	Sat, 3 Oct	R2 · Intermediate	3 presentations + discussion; brief Round 3	Groups 10–12
10	Sat, 10 Oct	R3 · Advanced	3 presentations + discussion; brief next groups	Groups 1–3
11	Sat, 17 Oct	R3 · Advanced	3 presentations + discussion; brief next groups	Groups 4–6
P1	Sat, 24 Oct	R4 · Project	Workshop — every group leaves with a suitable use case. On-site (online acceptable)	All groups
12	Sat, 31 Oct	R3 · Advanced	3 presentations + discussion; brief next groups	Groups 7–9
13	Sat, 7 Nov	R3 · Advanced	3 presentations + discussion (last Phase-1 week)	Groups 10–12
P2	Sat, 14 Nov	R4 · Project	Progress update — 20-min talk + 10-min Q&A	Groups 1–6
P3	Sat, 21 Nov	R4 · Project	Progress update — 20-min talk + 10-min Q&A	Groups 7–12
P4	Sat, 28 Nov	R4 · Project	Final pitch — 5-min pitch + 5-min Q&A, invited guests	All 12 groups

*Phase-1 group bookings are a template the instructor may rotate; each presenting group's topic is assigned in the previous week's final 30 minutes. The **project workshop is held on the 24 Oct holiday on-site with online attendance acceptable**, so it falls (by date) between Weeks 11 and 12, while Advanced topic weeks continue on 31 Oct and 7 Nov.*

Topic Batches

Each batch holds 12 topics — one per group per round. Groups pick from the released batch (first come, first served, no duplicates); the instructor confirms assignments and briefs them the week before.

Batch 1 — Basic (Round 1) [tentative]

1. LLMs and the Claude ecosystem — capabilities, limits, the model family, and choosing among API/SDK, Claude Cowork, and Claude Code.
2. Prompting fundamentals — instructions, context, examples, system prompts, XML tags, output formatting.
3. Reasoning and chain-of-thought — step-by-step prompting and when to use it.
4. The Messages API and SDK — first programmatic calls and parameters.
5. Streaming and structured/JSON output — producing validated machine-readable results.
6. Multimodal inputs and long documents — images, PDFs, and document question-answering.
7. Prompt evaluation and iteration — testing prompts and measuring quality with a rubric.
8. Tool use I — defining a tool and completing the tool-use loop.
9. Tool use II — multiple tools, handling results and errors.
10. Building a basic agent — the agent loop, goals, and stopping conditions.
11. Agent memory, context, and retrieval — conversation state, context limits, RAG basics.
12. Claude Code and Claude Cowork basics — Plan Mode, Explore→Plan→Code→Commit, and no-code file/task automation.

Batch 2 — Intermediate (Round 2) [tentative]

1. The Claude Code workflow in depth — Plan Mode and the Explore→Plan→Code→Test→Commit loop.
2. A lean CLAUDE.md and context budgeting.
3. Hooks, slash commands, and customizing Claude Code.
4. Skills and the marketplace — installing and using skills; subagents.
5. Claude Cowork for requirements — eliciting and specifying system requirements.
6. System design with agentic tools — architecture, data models, and diagrams.
7. Reading, reviewing, and verifying AI-generated code — the Do/Don't discipline.
8. Context engineering — turning a design document into agent context.
9. Agent specification I — SOUL.md and CLAUDE.md; identity, values, and boundaries.
10. Introduction to MCP — what it is and using existing connectors.
11. Retrieval in practice — building a small RAG pipeline for an agent.
12. Responsible use and prompt injection — hallucination, verification habits, and defence basics.

Batch 3 — Advanced (Round 3) [tentative]

1. Agent specification in depth — the SOUL.md 9-component model and memory policy.
2. Agent architectures — planning, reflection, and the ReAct pattern.
3. Advanced tool use and computer use.
4. MCP I — using MCP servers effectively.
5. MCP II — building and testing a custom MCP server.
6. Multi-agent systems I — orchestration, subagents, and delegation.
7. Multi-agent systems II — coordination and communication patterns.
8. Context engineering at scale — long-horizon context, memory, and compaction.
9. Retrieval at scale — advanced RAG, hybrid search, and reranking.
10. The Claude Agent SDK — building a production agent.
11. Authoring skills (SKILL.md) and custom coding agents.
12. Evaluating agents — eval design, benchmarks, and LLM-as-judge.

The Final Project (Round 4)

Round 4 replaces topic presentations with a **team project**: each group takes a real use case from idea to a pitched solution over four sessions, applying everything from Phase 1. It is worth **28 points**.

Session	Date	What happens	Format
Workshop	Sat, 24 Oct	Guided workshop so every group leaves with a suitable, well-scoped use case. (on-site; online acceptable)	Whole class; facilitated
Progress update A	Sat, 14 Nov	Groups 1–6 present progress; audience questions and discussion	20-min talk + 10-min Q&A
Progress update B	Sat, 21 Nov	Groups 7–12 present progress	20-min talk + 10-min Q&A
Final pitch	Sat, 28 Nov	All 12 groups pitch the finished project to invited guests	5-min pitch + 5-min Q&A

Project scoring (28 points)

Component	Points	Criteria
Use case (from the workshop)	5	A suitable, well-scoped, feasible use case with a clear problem and intended users.
Progress presentation	4	One 20-min progress update + Q&A: clear progress, working parts shown, sound plan to finish.
Progress discussion	$2 \times 2 = 4$	Individual discussion note in each of the two progress weeks (audience engagement); attendance required.
Final report	8	The written deliverable: working solution, evidence, testing/verification, agentic technique, AI utilization, impact, documentation, and integrity.
Final pitch	7	A persuasive 5-min pitch + 5-min Q&A to invited guests: framing, demo, impact, delivery, and Q&A.

Example project directions (groups may choose their own): a safety/guardrail layer for an agent; a deployed MCP-connected production agent; an applied multi-agent workflow; a domain assistant built by directing Claude Code; an evaluation/validation harness for a real task; or an impact-assessed AI solution for a real stakeholder.

Supporting documents (provided separately): *Final Project Workshop Guideline*, *Progress Report — Template*, *Final Report — Template*, and *Final Pitch Guideline*.

Assessment

The total score is out of **100 points**. There are **no examinations**. Points come from the Phase-1 deliverables and the Phase-2 final project. **Presentation and discussion require attendance: a student who is absent scores zero for that session's presentation and discussion note.**

Component	Unit	Count	Points	Total
Presentation — Document (3) + Delivery (4)	per round	3	7	21
Homework	per non-presenting week	9	3	27
Notes & discussion (<i>individual</i>)	per Phase-1 week	12	2	24
Final project — use case (5) + progress presentation (4) + progress discussion (2×2) + final pitch (15)	project	1	28	28
Total				100

Presentation and homework are **group work**; **notes and discussion are individual work**. In Phase 1 each group presents once per round (3 in all) and submits homework on the 9 weeks it is not presenting; every student submits discussion forms in all 12 Phase-1 weeks. The **final project (28)** is detailed under *The Final Project* above.

Scoring Criteria

Each Phase-1 deliverable is scored by the **checklists** below — one point per item ticked. Because difficulty rises each round, the bar for a tick rises with the level (Basic → Intermediate → Advanced); the **Level Expectations** table states what "complete and correct" means at each tier. (The final project is scored by the project table above.)

The shared document — Topic Practice Report

Both the presenting groups and the non-presenting groups submit the **same kind of document: a Topic Practice Report** (template provided separately as *2110576 Topic Practice Report — Template*). A presenting group submits one for its assigned topic; each non-presenting group submits one for each topic assigned to it as homework. The report is worth **3 points**, and the **same checklist** scores both the presentation **Document (3)** and the **Homework (3)**.

Document / Homework checklist (3 points — 1 per item)

✓	Pt	Criteria
☐	1	Complete & correct content — every template section filled; the topic is explained correctly; results/evidence are included and appropriate to the level.
☐	1	Critical reflection & verification — shows what was tried, what worked and what failed, how AI output was checked, and an honest statement of limits.
☐	1	Credit & disclosure — sources are real and correctly cited, AI use is disclosed, and the work is the group's own.

*Homework must be submitted **before class**; a report submitted after class starts, or copied, scores 0.*

Presentation delivery checklist (4 points — 1 per item; slides included)

Scores the live presentation and its slides. The items deliberately **do not repeat** the document criteria above — content, reflection, and citations are graded in the document, so delivery is graded on how it is presented. **A student absent on their group's presentation day scores 0 for this round's presentation, even if the rest of the group presents.**

✓	Pt	Criteria
☐	1	Slides — clear, well-structured slides that support the talk (visual, not walls of text).
☐	1	Live demonstration — runs the working result / demo live (or plays recorded evidence), not just describing it.
☐	1	Delivery & engagement — organized, clear, on time; all present members take part; presents so classmates can learn and the discussion is seeded.
☐	1	Q&A — answers questions from peers and the instructor competently.

Notes & Discussion (2 points, individual; per Phase-1 week)

Each student fills **three paper forms** — one per presentation that week — in English and short. Forms are **handed in at the end of class**, then **scanned and evaluated with Claude Cwork and double-checked by the lecturer**. Legible ("good enough") handwriting is required; new forms are always available. **Requires attendance.** (In the two progress-update weeks a student instead submits one progress-discussion note, scored under the project.)

Score	Criteria
2	Present; all three forms handed in and substantive — key points, a question, and a takeaway for each topic.
1	Present, but forms are thin, illegible, or one or more topics missing.
0	Absent (automatic zero), or no forms handed in.

Level Expectations — what full marks requires at each Phase-1 round

Level (Round)	Presentation & document	Homework	Discussion
Basic (R1)	A correct working demonstration of a foundational technique (a prompt, an API call, a basic tool or agent) with a short reflection.	Reproduce or attempt the basic technique for the three topics.	Capture the core concepts and raise one question per topic.
Intermediate (R2)	A working workflow that combines steps or tools (Claude Code workflow, context engineering, a small RAG or MCP connector).	Attempt the intermediate workflow and note the pitfalls.	Compare approaches and surface trade-offs.
Advanced (R3)	A non-trivial functioning system (custom MCP server, multi-agent orchestration, an Agent SDK build, an eval suite) with evidence it works.	Attempt the advanced build, or analyze its design.	Critique design choices and failure modes.

Grading Scale

A ≥ 80 · B+ 75–79 · B 70–74 · C+ 65–69 · C 60–64 · D+ 55–59 · D 50–54 · F < 50

Course Policies

Attendance. Attendance is checked in the first 30 minutes of every session. **A student who does not attend receives zero for that session's presentation and discussion note** — this includes a member of a presenting group who is absent on presentation day. Both are credited per individual student.

Deadlines. Homework must be submitted **before** the class in which its topics are presented. Discussion notes must be submitted **within the class day**. Presentations are delivered live in the booked week; a group that fails to present in its booked slot forfeits that presentation's points unless rescheduled in advance for a documented reason.

Groups and accounts. Groups of three are fixed for the semester and share one Claude account. All deliverables are submitted and graded per group. Groups are responsible for managing their shared account so it lasts the full term.

Academic Integrity. All submitted work must be the group's or student's own. Discussing concepts across groups is encouraged; copying another group's presentation, homework, or notes is treated as academic misconduct under Chulalongkorn University policy. Each presenting group's topic differs from every other group's in the same round, so submissions are expected to be distinct.

Not knowing is acceptable; not giving credit is not. Saying "we don't know," reporting a failed attempt, or presenting an incomplete result is fine and will not fail you — honesty about limits is expected. What is not acceptable is passing off others' work as your own or citing sources falsely.

Plagiarism and false citation are grounds for a failing grade (F) in the course. This applies to AI output as well: any content produced with Claude must be disclosed, and claimed sources, quotes, and results must be real and verifiable.

AI Tools & Critical Use. This is a course about agentic AI; students use Claude extensively. Students are expected to engage critically with AI output — verify claims, question assumptions, test results, and document their reasoning — not accept it passively. Any AI-assisted content must be disclosed and accompanied by the group's own analysis and reflection.